Is a Drawing Unit Just a Label? AutoCAD-Compatible Unit Scaling in GoSurvey's Traverse Editor and Clipboard

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Abstract

A Reddit report that the Traverse Closure Analysis panel labelled distances `(ft)` on a meters project led to a harder question: if a drawing's linear unit is just a label on a number, does changing it — or moving a number between two drawings with different labels — ever change anything real? This note answers both halves precisely. First, it confirms by inspection of GoSurvey's own code that changing a drawing's unit setting never rescales existing geometry, which is not a bug but the same convention AutoCAD's own `INSUNITS` system variable uses (Autodesk, 2026). Second, it finds and closes a real gap against that same convention: AutoCAD scales content pasted or inserted between two drawings of different units by the ratio of those units (Autodesk, 2026), but GoSurvey's clipboard paste skipped that check entirely whenever neither drawing belonged to one of its own "Projects". This note derives the governing unit-ratio formula, works a numeric example by hand, and documents the warn-then-convert implementation and its test suite — deliberately never silent, unlike AutoCAD's own default.

1 The question: is the unit label load-bearing?

GoSurvey stores every drawn length as a plain number with no unit attached — a line "100 units long" is stored as the double `100.0`, full stop. The drawing's `INSUNITS` code (Unitless, Inches, Feet, Millimeters or Meters; REQ-357) is a separate setting that says how that number should be read: 100 feet, or 100 meters, or something else entirely. Changing the setting after the fact does *not* walk back through the drawing and rescale every stored number to the new unit — the number `100.0` stays `100.0`, now read under the new label.

This is confirmed directly from the source rather than assumed: every write site for the drawing's unit setting (`src/ui/CadUiSettings.cpp`, `src/ui/CadUi_DrawingSettings.cpp`) assigns only the integer code itself (`cmd.drawingInsUnits = ...`); none of them touches a coordinate, a

distance or any entity array. No code path in the program rescales geometry in response to a unit-setting change.

This is deliberate, not an oversight, and it is the same rule AutoCAD follows. The Autodesk reference for `INSUNITS` describes it as the value AutoCAD uses to *auto-scale* content **being inserted or attached** — blocks, images, and xrefs — into a drawing; it does not describe `INSUNITS` as retroactively rescaling anything already drawn (Autodesk, 2026). Flipping the unit dropdown on an existing drawing, in both programs, only changes how future insertions and pastes are scaled and how existing numbers are labelled on screen — never what those numbers already are.

2 The real gap: paste between two differently-labelled drawings

If the label is informational for an existing drawing, the question moves to the boundary where two drawings meet: what happens when you copy an object out of a meters drawing and paste it into a feet drawing?

What AutoCAD does

AutoCAD computes the ratio of the source and destination drawings' `INSUNITS` values and scales the inserted or pasted content by that ratio, so that the object keeps its real-world size rather than its raw number (Autodesk, 2026). A millimeter-drawn object pasted into an inch-based drawing is scaled by 25.4, the millimeters-per-inch ratio, as a direct worked instance of the same rule (Autodesk, 2026). This scaling is automatic and, by default, silent: nothing stops to ask first. AutoCAD does expose a toggle for it (`DWGUNITS`'s "Scale objects from other drawings upon insert"), and user reports of surprising scale jumps are common enough that the standard community advice is simply to turn it off and keep every file in one unit (Autodesk, 2026).

What GoSurvey did, before this change

GoSurvey already replicated half of this correctly: inserting a BLOCK compares the block's own recorded unit against the current drawing's unit and computes a scale factor automatically (`CadBlockInsertUnitsScale`, gated by the existing REQ-357 "Scale inserted objects" toggle — GoSurvey's own equivalent of AutoCAD's `DWGUNITS` option).

Clipboard PASTE, however, did not. `CheckClipboardPaste` (`src/commands/ProjectWarnings.cpp`) returned immediately, with no unit check performed at all, whenever neither the source nor the destination drawing belonged to a GoSurvey *Project* (its own survey-point-sharing feature, unrelated to AutoCAD). REQ-383 clause 7 had documented this explicitly: "a paste between two standalone drawings is unchanged." Pasting a 100.0000-unit object from a meters drawing into a feet drawing landed it as 100.0000 feet — a real 228% oversize, silently, because no unit arithmetic ran at all.

3 The governing formula

GoSurvey already carries a meters-per-unit table for its five `INSUNITS` codes (`geo::MetersPerInsUnit`):

Code	Unit	Meters per unit
1	Inches	0.0254
2	Feet (international)	0.3048
2	Feet (U.S. survey)	$1200/3937 = 0.3048006096\ldots$
4	Millimeters	0.001
6	Meters	1.0

When only the unit differs (no map-projection zone is involved), the transform GoSurvey's existing conversion engine computes is a pure uniform scale about the world origin (GoSurvey, 2026):

$$k_{\text{unit}} \;=\; \frac{m_{\text{from}}}{m_{\text{to}}}, \tag{1}$$

where m_{from} and m_{to} are the source and destination drawings' meters-per-unit values (`geo::PlanConversion`, the `!plan.zoneDiffers` branch). Every pasted coordinate, length and height is multiplied by k_{unit}:

$$x_{\text{dest}} = k_{\text{unit}} \cdot x_{\text{src}}, \qquad y_{\text{dest}} = k_{\text{unit}} \cdot y_{\text{src}}, \qquad z_{\text{dest}} = k_{\text{unit}} \cdot z_{\text{src}}. \tag{2}$$

Equation 1 is exactly the same ratio AutoCAD computes for a cross-unit insert or paste (Autodesk, 2026); GoSurvey reuses its existing conversion engine rather than adding a second formula.

Worked example: meters into feet

A 100.0000-unit line, drawn in a meters drawing ($m_{\text{from}} = 1.0$), copied and pasted into a feet drawing ($m_{\text{to}} = 0.3048$, international foot):

Quantity	Formula	Value
Source length	drawn	100.0000 (m)
k_{unit}	$m_{\text{from}}/m_{\text{to}} = 1.0/0.3048$	3.2808398950
Pasted length, unconverted (before this change)	no scaling applied	100.0000 ft
Pasted length, converted (after this change)	$100.0000 \times k_{\text{unit}}$	328.0839895 ft

A 100 m line is 328.0839895 ft by definition (1 m = 1/0.3048 ft exactly); the converted figure matches to the number of decimals shown, confirming Equation 2 recovers the physically correct length rather than an approximation.

4 What this change does, and deliberately does not do

`CheckClipboardPaste` (`src/commands/ProjectWarnings.cpp`) now computes the unit mismatch check *before* the standalone-drawing early return, rather than inside the branch that early

return used to skip entirely. When both drawings are standalone and their units differ, the function returns `PasteCheck::Verdict::Warn` with an explanatory message and `canConvert` set by the same `PlanClipboardConversion` / `ApplyClipboardConversion` pair REQ-383 already uses for a Project-linked units mismatch — no second conversion path was written. The existing paste-prompt UI (`st.pastePrompt`) already handled a non-blocking warning with a convert option generically, so no new UI code was needed either.

Deliberately unchanged: a coordinate-system (map-projection zone) mismatch between two standalone drawings is still not checked. AutoCAD has no equivalent concept to check against for a plain drawing pairing, and extending that check would require deciding what a "zone" even means for two arbitrary standalone drawings — a materially different, unasked-for question from the one this change answers. Equation 1's scope is unit ratio only.

Deliberately not silent: AutoCAD applies this scaling without asking, which its own user community reports as a frequent source of surprise (Autodesk, 2026). GoSurvey's version always surfaces the warning and lets the user choose *Convert* or *Cancel* — replicating the arithmetic, not the silence.

5 Automated test summary

File	Test case	What it proves
`ProjectWarningsTests.cpp`	two standalone drawings and the same project are never checked	a ZONE-only mismatch between standalone drawings still returns `Ok` (unchanged)
`ProjectWarningsTests.cpp`	two standalone drawings with a unit mismatch still warn and offer convert (issue #831)	a UNIT-only mismatch between standalone drawings now returns `Warn` with `canConvert`, and `StartPasteCommand` opens a non-blocking prompt
`ProjectWarningsTests.cpp`	a units mismatch warns / can be converted: lengths, heights, radii and the origin all follow	pre-existing Project-linked path is unchanged by this edit

All cases pass under the `GoSurveySnapTests` executable (Catch2, MSVC release build); the four unrelated pre-existing `[req383]` failures in this build (point-visibility renumbering, tracked separately) are identical before and after this change, confirmed by running the same filtered suite against both revisions. The source is PR #837, branch `fix/issue831-traverse-closure-units`, amending REQ-383 clause 7 in `spec/requirements.md` (decision `D-2026-10-11-c` in `spec/project.md`).

References

Autodesk, Inc. (retrieved 2026). *INSUNITS (System Variable)*, AutoCAD / AutoCAD LT Help. Autodesk Knowledge Network. `https://help.autodesk.com/cloudhelp/2022/ENU/AutoCAD-LT-MAC/files/GUID-A58A87BB-482B-4042-A00A-EEF55A2B4FD8.htm`

GoSurvey Project (2026). *Source repository*, functions `geo::MetersPerInsUnit`, `geo::PlanConversion` (`src/geo/DrawingConversion.cpp`), `CheckClipboardPaste` (`src/commands/ProjectWarnings.cpp`), and `CadBlockInsertUnitsScale` (`src/commands/CadBlocks.cpp`). `https://github.com/chetjones003/GoSurvey`